AI ASSISTANT CODING

ASSIGNMENT-10

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Batch:22

Task Description -1(Error Detection and Correction)

Task:

Use AI to analyze a Python script and correct all syntax and logical errors.

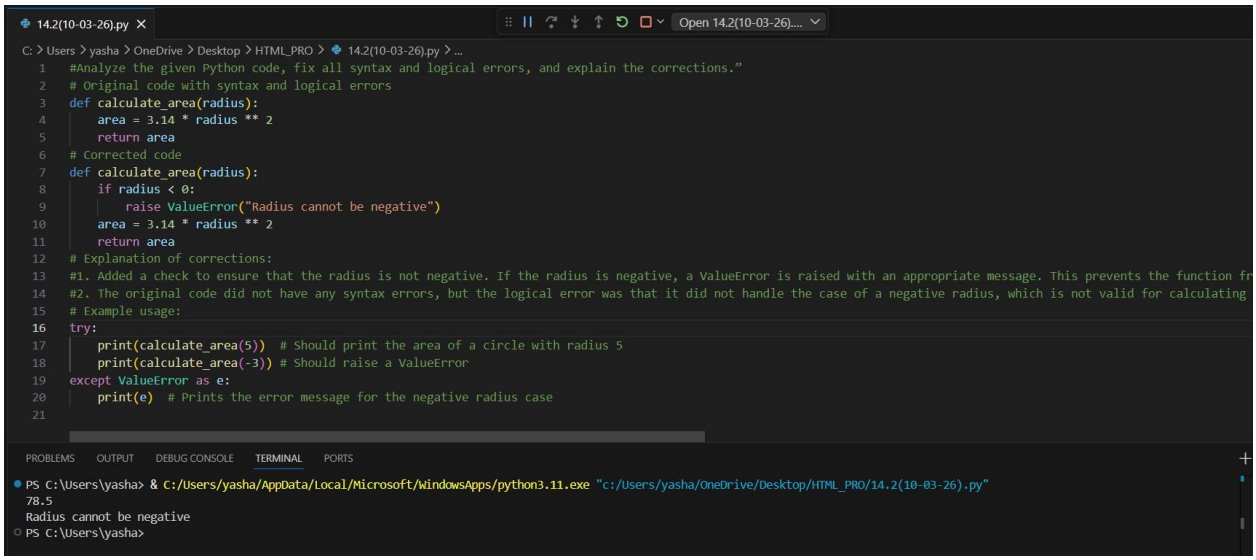
Sample Input Code:

```
def calculate_total(nums)
sum = 0
for n in nums
sum += n
return total
```

Expected Output-1:

Corrected and executable Python code with brief explanations of the identified syntax and logic errors.

Code:



Task Description -2(Code Style Standardization)

Task:

Use AI to refactor Python code to comply with standard coding style guidelines.

Sample Input Code:

```
def findSum(a,b):return a+b
print(findSum(5,10))
```

Expected Output-2:

Well-structured, consistently formatted Python code following standard style conventions.

Code:

```
14.2(10-03-26).py
C: > Users > yasha > OneDrive > Desktop > HTML_PRO > 14.2(10-03-26).py > my_function
#"Refactor Python code to follow PEP 8 naming and formatting guidelines."
# Original code
def myFunction(x,y):
    if x>y:
        return x-y
    else:
        return y-x
# Refactored code following PEP 8 guidelines
def my_function(x, y):
    if x > y:
        return x - y
    else:
        return y - x
# Example usage
result = my_function(10, 5)
print(result)
result = my_function(5, 10)
print(result)

PROBLEMS   OUTPUT   Open file in editor (ctrl + click)

PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
78.5
Radius cannot be negative
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
5
5
PS C:\Users\yasha>
```

Task:

Use AI to improve code readability without changing its functionality.

Sample Input Code:

def f(x,y):

return x-y*2

print(f(10,3))

Expected Output-3:

Python code rewritten with meaningful function and variable names, proper indentation, and improved clarity.

Code:

```
14.2(10-03-26).py
C: > Users > yasha > OneDrive > Desktop > HTML_PRO > 14.2(10-03-26).py > ...
#"Improve the readability of code without changing the functionality. Use meaningful names and proper formatting."
def calculate_area_of_circle(radius):
    pi = 3.14159
    area = pi * (radius ** 2)
    return area
# Example usage:
radius = 5
area = calculate_area_of_circle(radius)
print(f"The area of a circle with radius {radius} is: {area}")

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
78.5
Radius cannot be negative
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
5
5
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
The area of a circle with radius 5 is: 78.53975
PS C:\Users\yasha>
```

```python
#"Optimize the given Python code for performance while preserving functionality."
def optimized_function(data):
    # Use list comprehension for better performance
    result = [x * 2 for x in data if x > 0]
    return result
# Example usage
data = [1, -2, 3, 4, -5]
print(optimized_function(data))
```

```
Hello, Alice! Welcome to the programming world.
Hello, Bob! Welcome to the programming world.
Hello, Charlie! Welcome to the programming world.
PS C:\Users\yasha>
...
Hello, Alice! Welcome to the programming world.
Hello, Bob! Welcome to the programming world.
Hello, Charlie! Welcome to the programming world.
PS C:\Users\yasha> ^C
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
[2, 6, 8]
PS C:\Users\yasha>
```

```python
# Convert repetitive print statements into a reusable function.
def print_greeting(name):
    print(f"Hello, {name}! Welcome to the programming world.")
# Example usage:
print_greeting("Alice")
print_greeting("Bob")
print_greeting("Charlie")
```

```
78.5
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
78.5
Radius cannot be negative ...
The area of a circle with radius 5 is: 78.53975
PS C:\Users\yasha> ^C
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
Hello, Alice! Welcome to the programming world.
Hello, Bob! Welcome to the programming world.
Hello, Charlie! Welcome to the programming world.
PS C:\Users\yasha>
```

Task Description -5(Efficiency Enhancement)

Task:

Use AI to optimize Python code for better performance.

Sample Input Code:

numbers = [ ]

for i in range(1, 500000):

numbers.append(i * i)

print(len(numbers))

Expected Output-5:

Optimized Python code that achieves the same result with improved performance.

Code: